

Transformative Outcomes, Engineering Policy and Engineering Education: Strengthening the Impact of TE on Socially-Relevant Problems

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Abstract. Engineering in the coming decades need to address wicked problems of environmental sustainability and social justice. Engineers and engineering practice could enable major transformations in the systems of basic service provision, such as energy, mobility, food, housing and healthcare. Transdisciplinary engineering (TE) provides alternatives for reconfiguring engineering with the demands of meeting these wicked problems, many of which have been created or exacerbated with traditional engineering. Supporting approaches from wider society include engineering policy and engineering education. But how can we judge if TE is supporting the reconfiguration of engineering effectively? And how can we identify actions within engineering policy and engineering education that could support this? To understand this, we explore the use of the Transformative Outcomes framework (TOF) to appraise progress and identify possible actions to support TE in the UK. This treats TE as a niche innovation within the engineering landscape, that has to develop in the face of the traditional engineering regime. Focusing on four outcomes from the TOF that cover both niche and regime level: shielding, upscaling, de-aligning and unlearning, we provisionally identify actions in place in the UK and possible gaps where UK policy and education for engineering could go further in supporting TE's move into a new engineering regime. For engineering policy and education, TOF generates classes of possible action that can enable alignment across wider-economy and within training and skills. This provides the basis for a wider and more systematic application of the TOF for developing TE in the UK and internationally.

Keywords. Transdisciplinary engineering, transformative outcomes, engineering education, engineering policy, socially-relevant

Introduction

Societies are facing many interconnected existential threats – from climate and biodiversity loss, human health and disease outbreaks to international conflict. More than ever, societies are going to become increasingly reliant on engineering systems – not just the current ones in place, but innovative ones required to address these future threats. Within this context, Transdisciplinary Engineering (TE) has the potential to be an important way of ensuring, as Cooper et al [1] put it, that we “get the engineering we need, not just the engineering we have”. TE, an emerging practice in engineering

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research, follows similar goals to wider transdisciplinary movements in sustainability and health research.

These goals were set out in one of the founding papers on TE [2] which highlights the role of TE in tackling ‘ill-defined’ and ‘socially-relevant issues’. This orientation aligns well with other key papers from the wider T community, e.g. [3] which defines transdisciplinary research (TR) as having 3 components:

- a) Focusing on societally relevant problems
- b) Enabling mutual learning process from different disciplines and outside academia (i.e. collaboration across academic disciplines and outside academia)
- c) Creating robust knowledge that is solution-oriented socially robust and transferable to both the scientific and societal practice

Despite TE having a significant overlap with wider TR approach, at a conceptual level at least, there has been limited convergence in the content of research between these two fields. As Cooper [4] has shown, TE research forms a distinct and segregated field of study suggesting that TE is driven more by what we might call normal engineering research interests than a broader transformative agenda that transdisciplinarity entails. In his paper he showed that TE research was principally concerned with “‘industrial engineering, ‘commerce’ and ‘product development’” (p.918). This sits in contrast with the TR research which was characterised with “‘participation’, ‘stakeholder involvement’, ‘co-production’ and ‘action research’” (p.918).

This suggests that while TE represents an important shift in engineering research practice that places greater emphasis on being more socially-relevant, there is still a gap in the research capacity of the field to translate that into a significant orientation towards being more ‘socially-relevant’ such as the role of transdisciplinary engineering in healthcare, food provision and housing, beyond the world of industrial product design and manufacturing. If TE is to play a role in enabling the wider engineering community to support the transition to clean energy in an effective, rapid and just way, then we need to consider how TE scales up into the mainstream of engineering practice.

This of course raises the question of how we might understand what kind of support is necessary to enable emerging research and industry practice like TE to become a more standard, mainstream way of acting. To address this question of how, we draw on a research field outside of engineering, but related to understanding how novel engineering and technology innovations emerge – the field of innovation management. In particular, we draw on an approach known as Transformative Innovation Policy (e.g. [5, 6]) which is particularly suited to the ideas presented here: how novel approaches can support a just transition – that is to say, address socially-relevant challenges. Within this approach, scholars have developed a framework that helps identify approaches which can support transformative innovations move from small-scale activities to more widely adopted mainstream approaches: the Transformative Outcomes Framework (TOF) [7]. We use the TOF as a checklist to explore how TE can be supported to grow, with two other lenses: engineering education and engineering policy. While engineering education is a widely developed field of academic inquiry and scholarship reflecting insights about the kinds of approaches to teaching engineering which of course sets direction for the kind of engineering a country has, engineering policy is much less developed.

This paper is structured as follows: Section 1 introduces the Transformative Innovation Policy (abbreviated to TIP) and Transformative outcomes framework (abbreviated to TOF). In section 2, we discuss engineering policy (abbreviated to EP) and engineering education (abbreviated to EE) as ways of shaping engineering. Section 3 explores the potential for EE and EP to support TE development by applying the TOF, illustrating activities and gaps in EE and EP in the UK. In Section 4, we draw conclusions summarising the implications of this research for TOF as well as for EE and EP in UK communities.

1. Transformative Innovation Policy and Transformative Outcomes Framework

1.1 Transformative Innovation Policy - background

As noted above, Transformative Outcomes framework (TOF) is developed in the field of Transformative Innovation Policy (TIP) research [6]. This interdisciplinary research community is interested in studying and implementing holistic change in how people and the planet could thrive addressing interconnected and wicked challenges of unsustainable growth, environmental degradation, income and gender inequalities, extractivism and insecurities. In this context, TOF was first developed by Ghosh et al [7], to guide actions towards transformative change. At the heart of this framework is a theory of socio-technical change. This theory suggests that transforming energy, health or mobility systems in a city, region or a country require dedicated support in developing and mainstreaming alternative and innovative configurations of the technological, economic and social (known as ‘niches’). For these niches to become dominant, the incumbent socio-technical configuration (also known as the ‘regime’) needs to open up, be less resistant to change and create space for alternatives to flourish. Transformative change happens at the intersection and combination of these two dynamics: on one hand development nurturing and mainstreaming of niche, and on the other hand destabilisation, phase out or reconfiguration of the incumbent regimes [6].

There are 12 TOs proposed by Ghosh et al [7] - 4 for each of the 3 ‘macro-processes’: 1) Nurturing the niche, 2) Mainstreaming the niche, 3) Opening up the regime. In recent years the framework is further expanded and adjusted to include multi-system dynamics, unintended consequences of transition action and social justice considerations ([8, 9]). The framework is widely used by policymakers and investors to reorient their strategies, visions and portfolios and measure the real impact of their actions towards more socially just and environmentally sustainable directions.

We use the TO framework to explore what it means for engineering to contribute to socially-relevant challenges in a holistic way. This opens up a set of goals for understanding the wider societal system that supports the kind of engineering a country has (its engineering policy) and the approach it takes to generating development of engineering (engineering education). While we don’t explore this directly here, it opens up the question of what happens to the ‘normal’ or mainstream engineering practice, should TE grow from being a niche alternative into a more widely adopted approach (i.e

the regime). While TIP literature has many ways of characterising the shifting character of the ‘regime-level’ practices (from ‘creative destruction’ [8], through ‘decline’ and ‘phase out’ [10] we prefer to characterise the future engineering being ‘reconfigured’ in line with Laakso et al [11], reflecting the notion that “that several institutions, actors, practices, and constituent elements of practices are expected to change ..., resulting in a new combination of old and new elements”([11] p.16). We now turn to TOF that provides insight into how TE can be developed as a ‘transformative innovation’ in engineering.

1.2 The Transformative Innovation Framework

The TOF provides a set of conditions that would allow TE community to see where to intervene in order to transform engineering for a better future, based on the niche-regime theory of change (described in section 1.1). The framework is adopted from TIP literature which could be considered a close ally to the engineering policy community. The framework is essentially a set of outcomes that engineering actors could try to achieve, to ensure that their efforts are truly transforming the dysfunctional aspects of the engineering regime. Shifting entire regimes is way more complex and long term than addressing localised problems in the short term. E.g. An existing problem of congestion in cities can be solved by constructing new highways. But this solution doesn’t address the deeper issues such as why people in this city are buying cars in the first place? Have previously built highways incentivising more people to commute by cars? Could the issue be resolved by improving public transport or making distant neighborhoods accessible by mass transit systems? Congestion is therefore a wicked problem that requires engineering to do more than providing infrastructure solutions to the immediate problem. Transformations entail a) addressing problems from previous technofixes and learning from it, b) designing and implementing more systemic solutions (considering policies, politics, social, economic and cultural factors) rather than just technologies and infrastructures.

2. Engineering education and engineering policy in support of TE

Here we look at two key levers that shape the way engineering is developed as a country-level resource: engineering education and the wider concept of engineering policy. We first look at how engineering education plays a role in defining and developing engineering, as a precursor to exploring the role it has played in the UK for developing and supporting TE at the niche level, and reconfiguring mainstream engineering at the regime level. We then turn to explore the role of engineering policy – understood as a corollary to science policy, rather than as an umbrella term for ‘policies that clearly require engineering, such as transport, infrastructure and energy policy’.

2.1 Engineering education as a way of shaping engineering

Traditional Engineering education develops technical, design, and problem-solving skills through theoretical learning and practical application for real-world innovation [12,

13], and it integrates scientific and engineering principles to develop products and processes that contribute to societal advancement [14]. Building on this foundation, transdisciplinary engineering education is evolving to address complex global challenges by fostering collaboration between engineers and non-engineers, including local communities within university spaces [15]. This approach aims to create a more inclusive and decolonial curriculum, recognizing the diverse perspectives and knowledge systems that contribute to engineering solutions [16], with significant potential for positive transformation.

2.2 Engineering policy as a way of shaping engineering

Engineering policy is a term that is infrequently used in the academic research compared with the widely adopted, and closely related term ‘science policy’. Science policy generally refers to the approach a government will take in supporting and therefore shaping the kind and volume of science capability in a country. This typically includes how funding is set up, the goals of science, what makes for good science and so on. We can apply this same style of policy thinking to engineering as Cooper et al did [1]. This forces us to consider what kinds of support – funding, regulation, explicit goals or targets set by what institutions - engineering in a particular country has to develop in a particular way. We apply this thinking to TE via the TOF alongside EE in the next section.

3. Exploring the potential for EE and EP to support TE development

As noted earlier, the original TOF has a large number of outcomes cutting across both the niche and regime levels of analysis. Given the limited space for analysis in this paper we will explore how UK engineering education approaches (EE) and engineering policy have supported TE or where there are obvious steps that could be taken to support it further. At this stage, this analysis is a proof-of-concept to explore, demonstrate and critique the applicability of the TOF for reflecting on the progress and support of TE as a niche innovation engineering research practice. Therefore, at this stage, the elaboration of activities associated with each of the outcomes used in this paper against EE or EP are selective, based on the authors insights and experience in the UK. Such work sets the stage for further developing the process into a more systematic appraisal of EE and EP actions for any country, as well as a reflection on the pros and cons of the TOF itself for this purpose

In Table 1, we explore how specific elements, approaches, aspects to EP and EE do or could help achieve each of the transformative outcomes in the case of the UK, a setting known to the authors. We note that the examples used in Table 1 are based on the authors’ work and professional exposure. Outline method for selecting outcomes and identifying supportive actions

Each of the four outcomes, namely 1) Shielding; 2) Upscaling; 3) Dealigning and 4) Unlearning. fall into one of the two domains of the TOF that refer either to the ‘niche’ level or ‘regime’ level. Our principal concern with selecting these outcomes was to keep the number to four and to split them evenly across niche and regime level. These four likely represent the easier outcomes to apply in this setting, so it is possible that extending this analysis to all twelve outcomes will reveal different findings than those discussed here.

Outcome	Definition	What this means for TE	Role of Engineering Policy	Role of Engineering Education
Niche-level				
Shielding	Protection for niche innovations and normalising these protection measures	Protection for TE as a niche alternative to conventional engineering and normalising of protection measures (i.e not a one-off intervention, but more permanent and stable means of ensuring TE exists) eg. the very existence of the TE conference, ISTE community and publications can be considered as means of protecting this community.	What UK is doing: • The UK's Royal Academy of Engineering represents a shield for TE as its programmes and priorities align with TE goals. • The focus on engineering ethics in the UK's Engineering Council, helps shield TE. What UK could do: • Set up a special funded programme for scholars operating under TE approaches • Support international network with the World Engineering Council	What UK is doing: • Creating space for TE through aligned teaching and research agendas • Eg. Bsc Science and Engineering for Social Change • UCL Education for Sustainable Development supporting environmental sustainability in engineering • Integration of Equality, Diversity and Inclusion (EDI) principles in engineering curriculum. What UK could do: • Invest in staff and student recruitment in the TE space • Further promote TE education
Upscaling	Increasing adoption of innovation by users and emerging signs of the stabilisation of new preferences	Increasing adoption of TE within the engineering community, observed through evidence of network expansion over the years, or wider uptake of TE ideas, or impact on wider society. For upscaling to be transformative, the effect needs to sustain.	What UK is doing: • The UK's Government Science and Engineering profession supports promotional activities² What UK could do: • Improve engagement between engineers and non-engineers in government • Develop open innovation institutes	What UK is doing: • New models and institutions for EE (NMiTE, TEDI London, Dyson Institute) • Increase in TE-centred programmes (e.g. UCL BSc Science and Engineering for Social Change [17], Warwick's MSc Humanitarian Engineering³) What UK could do: • Encourage staff and students' affiliation with institution promoting TE (i.e ISTE)
Regime-level				

² See: <https://www.gov.uk/government/news/what-does-the-government-science-engineering-gse-profession-offer-everyone>

³ See: <https://warwick.ac.uk/fac/sci/eng/postgraduate/humanitarianengineering/>

Dealigning	Confronting the mismatch between existing technologies, policies and social needs, a first sign of weakening of position of incumbent system	Confronting the dysfunctional nature of engineering solutions and practises (eg technofixes and male dominated disciplines) and highlighting the problematic nature of it. It is transformative when de-alignment - a deliberate action - is legitimised by a large TE community i.e. community actively disrupts the stability of traditional engineering.	What the UK is doing: • RAEng's report on Engineers 2030 [18] • The UKRI programme: Tomorrow's Engineering Research Challenges⁴ What the UK could do: • Reorient public organisations (e.g. Faraday Institution) away from normal engineering • Align infrastructure commissioning with TE	What the UK is doing: • Commissioning skills reviews (RAEng) • Redefining professional accreditation criteria (Engineering Council) What the UK could do: • Further update graduate profiles • Promote pre-university education cross disciplinary alignment
Unlearning	Abandoning some existing practices, absorbing risks of changing practices, internalising costs of reorganisation	Current engineering community abandoning some of the existing practises, beliefs, ideas and viewpoints. They are questioning assumptions underlying something they knew as normal before (e.g. not considering environmental impact of manufacturing processes upfront). It is transformative when there is a self-reflexivity at play at large among the next generation of engineers.	What the UK is doing: • ICE director talking about a new way of doing engineering [19] • Major engineering organisation eg. National Highways adopting a new Sustainability director What UK could do: • Promotion via novel pathways across engineering/encourage policy/engineering/academy interchange • Implementation of Industry 5.0 into the wider workforce and supply chains	What the UK is doing: • Adopt PBL, PBjL, CBL etc enabling transdisciplinary collaboration widely • Embedding novel curricula (in existing/new transdisciplinary programmes) • Forstering industrial collaboration (Industrial Advisory Boards) What UK could do: • Allocate further resource to transdisciplinary portfolio expansion • Reduce university disciplinary boundaries • Invest in staff CPD and retraining

Table 1. UK's actions and opportunities in engineering policy and education across the 4 transformative outcomes.

⁴ See: <https://www.ukri.org/publications/tomorrows-engineering-research-challenges/>

For each outcome, the definition according to the original TIP literature is presented alongside an interpretation for what that means for the engineering context in particular. This reflects that the application of the TOF is at once novel, but also meta-recursive: it is applying a framework about transformational innovation to a national resource (engineering) that is central to innovation (hence meta-level) to shape that resource to better transformational outcomes (hence recursive). Then we identified obvious signs or activity in the UK that match the engineering-oriented definition of the outcome to illustrate activity in support of TE. As noted, this was not a systematic appraisal of possible activity across say, public, non-governmental and private sectors, but simply an initial assessment of whether such activities are identifiable – that is, is it possible to locate activities at all? Future analysis will allow for a deeper, more systematic approach.

3.1 Initial results regarding supportive EE and EP actions for TE development

We can see that actions associated with **engineering policy** (EP) are identifiable for all the outcomes, with possibly ‘Upscaling’ (niche level) being the one with the most limited actions associated with it. This is perhaps reflected in the wider slow emergence of TE in the UK community. To illustrate this, we can observe publications using TE concepts (either the narrow/specific phrase ‘transdisciplinary engineering’ or the more inclusive/wider combination of both ‘transdisciplinary’ and ‘engineer*’ together in the same paper) with authors at UK-affiliated institutions on a major abstracting database like Scopus. We find that there are only 53 publications using the TE phrase with UK-affiliated authors, which expands to 178 with the wider search (which, without the UK element, returns 1,682 – therefore around 10% of TE-related research is UK-affiliated). This represents a small portion of the overall TE-related research globally, reflecting how much TE can develop in the UK.

Similarly, **engineering education** (EE) action is identified in relation to all dimensions, though many times insular rather than being the status-quo. Change can be slow, as UK institutions are risk averse in the present financial environment. While postgraduate provision can be more agile and requires less investment, undergraduate provision is slowly catching up. An added complexity here is that pre-university education is mainly disciplinary, meaning that embarking on a transdisciplinary undergraduate programme requires a cultural shift, with students often reporting that they want more depth in technical expertise instead of breath of interdisciplinarity, as that is the model that they are familiar with. However, the work done by the RAEng and the Engineering Council incentivizes institutions to reframe their curricula and graduate profiles, and existing engineering programmes are updated to embed a degree of transdisciplinary knowledge which is also beneficial. Sustained investment in programme development, close collaboration with industry and an aligned marketing and recruitment strategy will undoubtedly consolidate success.

4. Conclusions

4.1 Implications for TOF

This meta-recursive analysis verifies that TOF applied for the first time outside of innovation policy, is still a relevant tool to assess transformative potential for

Engineering to address wider societal challenges. The analysis highlights that Upscaling a niche and sustaining at scale could be a challenge, despite widespread acknowledgement of the benefit of the niche. In contrast to TIP where regimes are proven to be difficult to destabilise, in the Engineering space, outcomes related to ‘opening up of the regime’ namely de-aligning and unlearning are perhaps being achieved at a more desirable scale and pace. De-aligning can seemingly be done through ‘reconfiguration’ activities in policy and education settings. This supports ongoing debate in the sustainability transitions field about pathways of transformation: whether gradual reconfiguration of regimes is a more practical ambition to pursue, rather than radical shifts and substitution of the old by a new regime.

4.2 Implications for engineering policy and engineering education

Our initial analysis using the TOF reveals both progress and possibility for UK engineering policy and education. We see how key institutions are creating conditions for a regime reconfiguration by promoting novel ideas and standards that align with concepts and emerging practice in TE. What is interesting about the outcome of using the TOF is how it generates certain classes of possible action in a way that can enable alignment across wider-economy approaches (engineering policy) and within the training and skills sector (engineering education) that have the potential to be mutually reinforcing. With further analysis and a wider, more systematic survey of actions, coupled with a deeper appraisal of the impact and reach of those actions and input from practising engineers and current engineering students, it would be possible to have a clearer idea of how real the progress is made, and how much more effort is required to move TE into mainstream engineering practice.

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